

Università Bocconi

Macroeconomics

(Economics – Module 2 -30066)

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Chapter 19:

The Goods Market

in an Open Economy

The IS Curve in an Open Economy

In an open economy:

domestic demand for goods $\neq$ demand for domestic goods



Value of demand by citizens of
a country
(some of this demand falls on
foreign goods)



Value of demand for goods
produced in the country
(some of this demand comes from
foreigners)

In a *closed* economy, the 2 are equal:

$$Z = C + I + G$$

In an *open* economy, not equal anymore!

As policymakers of a country, which one do we care about most?

The Demand for Domestic Goods

$$Z = C + I + G - IM/\varepsilon + X$$

Domestic demand for goods

NX=Net exports

- Z = Demand for domestic goods
- X = Exports (foreign demand for domestic goods)
- IM = Imports (Domestic demand for foreign goods, in terms of foreign goods)
- ε = Real exchange rate (price of domestic goods in terms of foreign goods)
- IM/ε = Imports (in terms of domestic goods)
- Assume that C depends on $(Y-T)$ and I depends on Y and r (for now, forget about x and about expectations, for simplicity)

Two new terms: X and IM

What does IM depend on?

- An increase in **domestic income** will increase imports (high Y, higher demand for domestic AND foreign goods).
- An appreciation of the **real exchange rate** will increase imports (foreign goods are less expensive, since less domestic currency needed to buy foreign currency).

$$IM = IM(Y, \varepsilon)$$

(+ +)

- Note: an increase in ε increases IM, but $1 / \varepsilon$ goes down:
 - Demand for IM goes up
 - Value of imports up or down? Ambiguous

Two new terms: X and IM

What does X depend on?

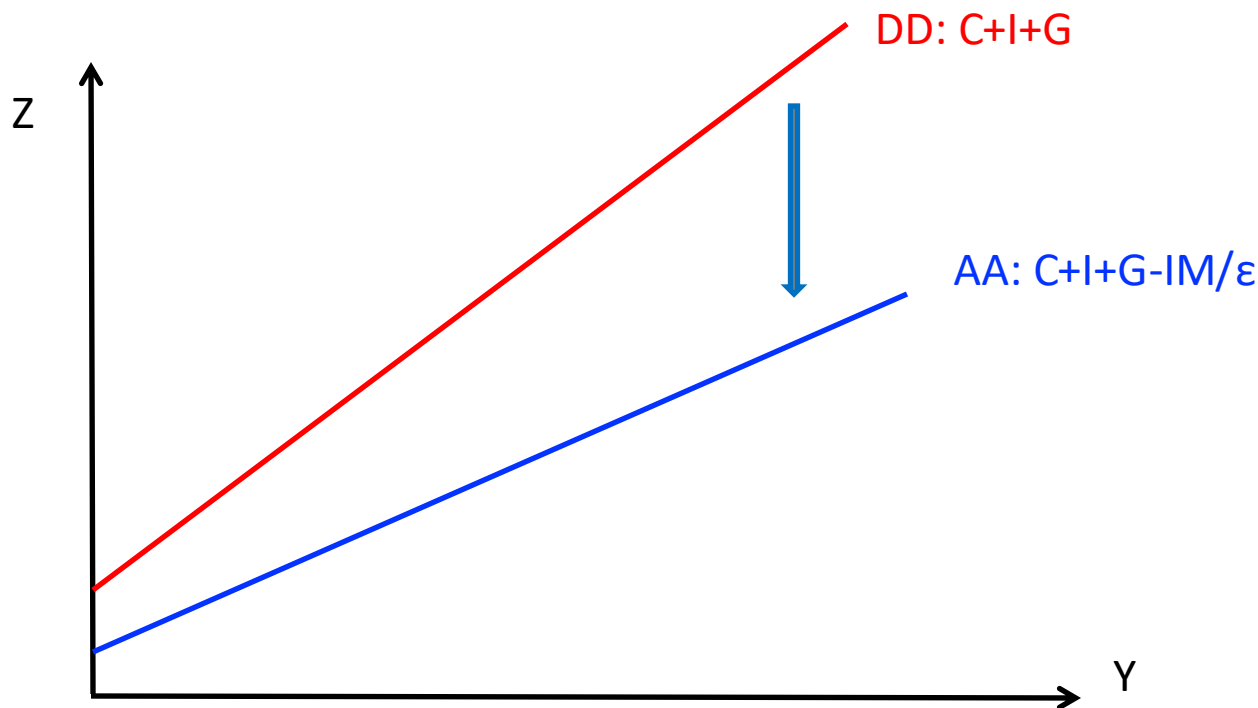
- An increase in **foreign income** will increase exports (high Y^* , higher demand for my goods from the foreign country).
- A depreciation of the **real exchange rate** will increase exports (domestic goods are less expensive, since less foreign currency needed to buy domestic currency).

$$X = X(Y^* , \varepsilon)$$

(+ -)

Towards the new Z Curve – step 1

1. from old Z we subtract $IM(Y, \epsilon)/\epsilon$, which depends on Y (the larger Y , the larger IM)

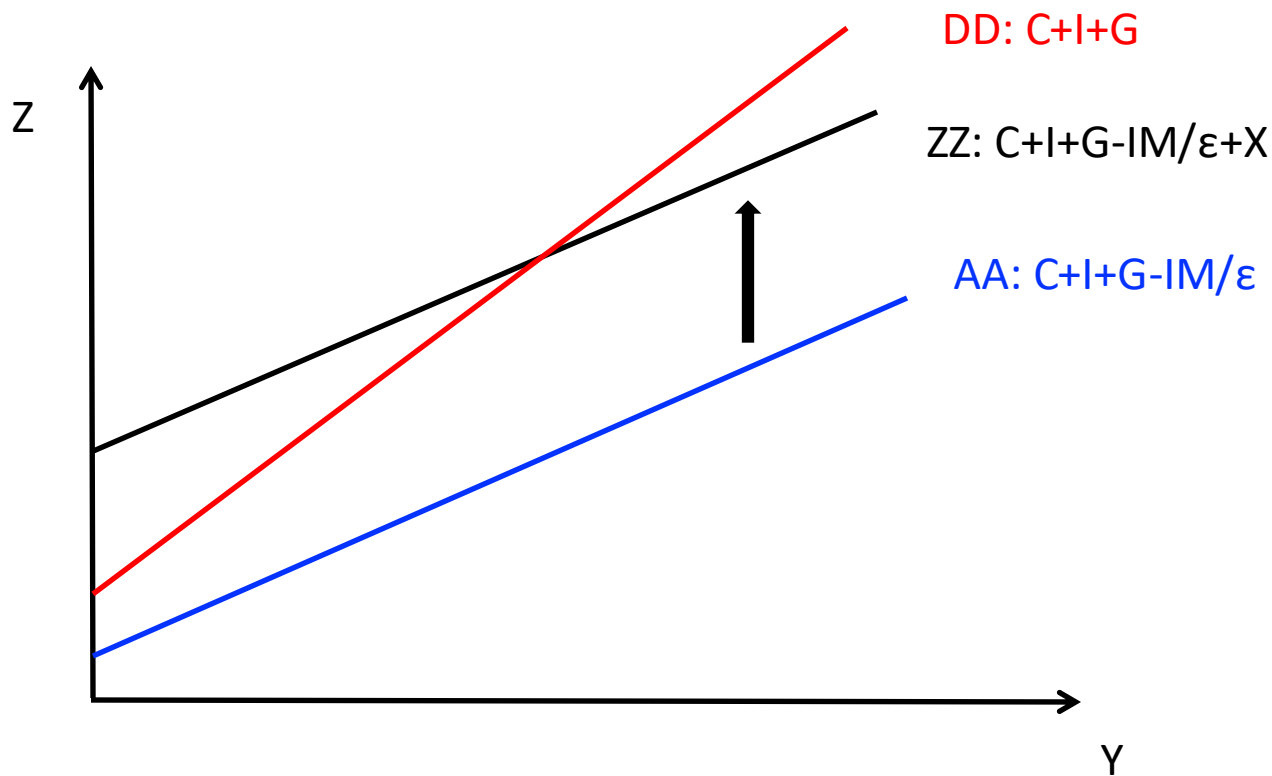


Observations:

- As for DD, AA slopes up, and slope of $AA < 1$: $Y \uparrow$, demand $\uparrow$ by less than 1
- AA flatter than DD: as $Y \uparrow$, part of higher demand goes to imports (but still sloping up, as long as some of the higher demand goes to domestic goods)

Towards the new Z Curve – step 2

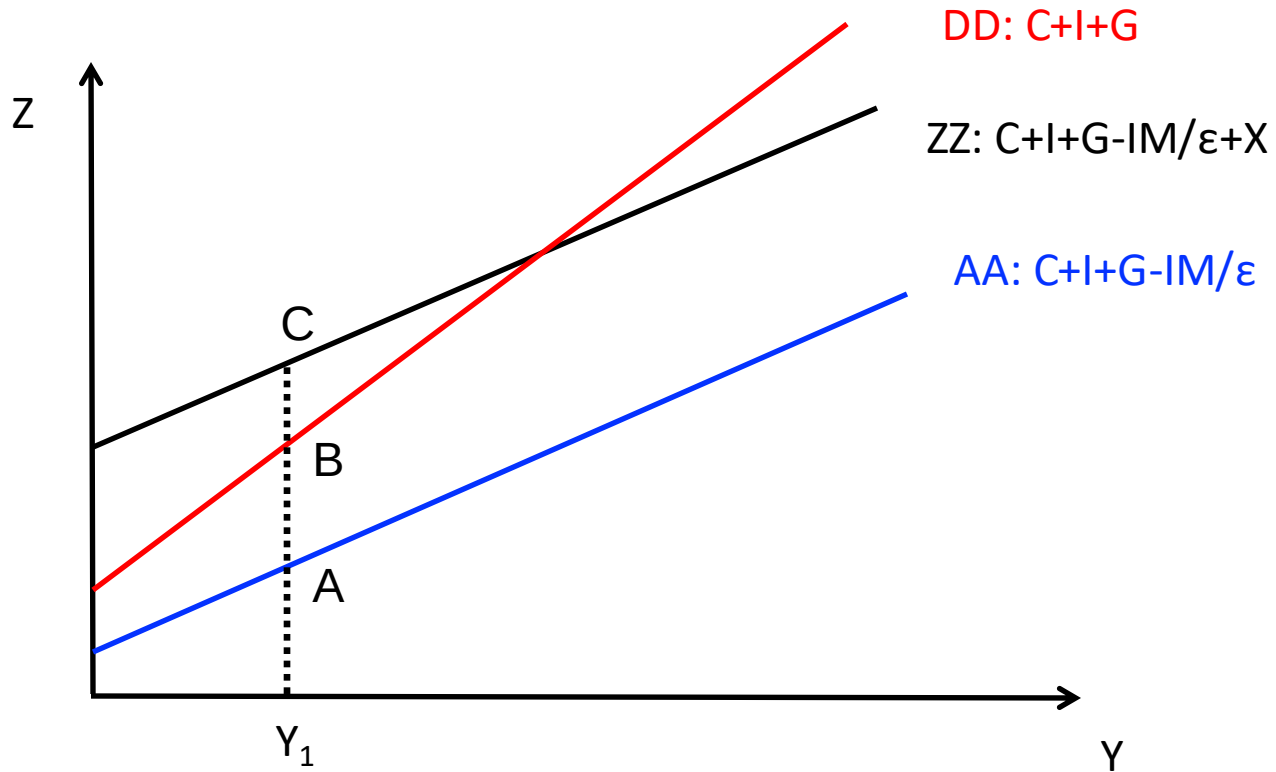
2. To this new curve we add X , which does not depend on Y



ZZ gives the Demand for Domestic Goods as a function of Y

Net Exports Graphically

If output is Y_1 , then what are values of exports and imports?

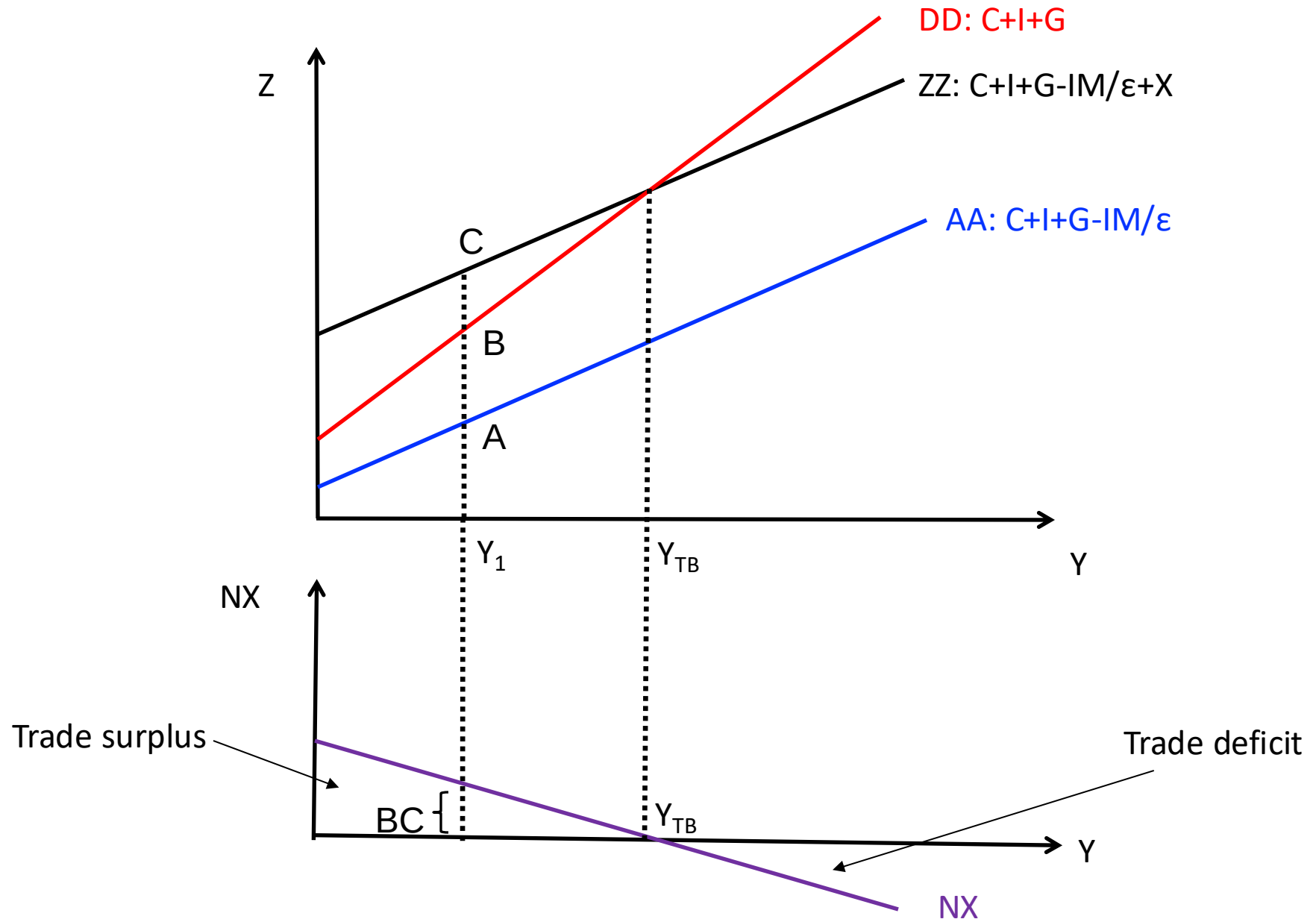


Exports are the distance AC

Imports are the distance AB

Net exports are the distance BC (in this case, positive as $X > IM/\epsilon$)

Net Exports Graphically

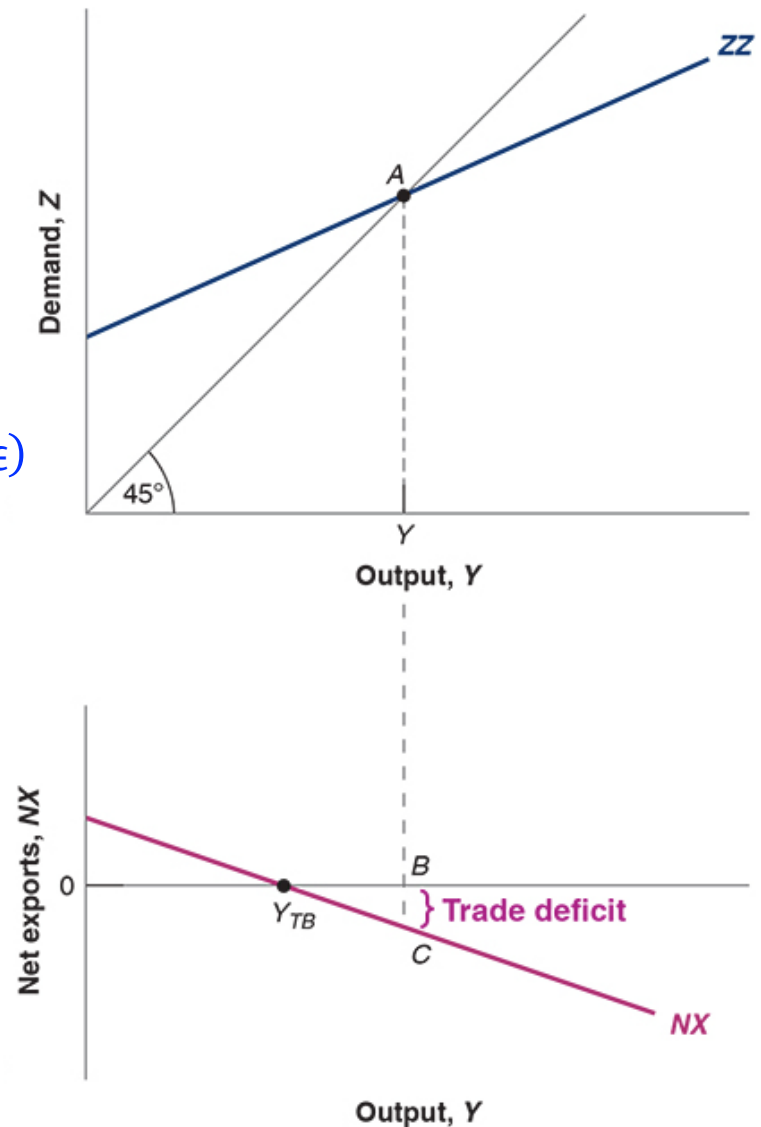


Trade balance, trade deficit or trade surplus

- Net exports decreasing with income
- If domestic income is equal to Y_{TB} then the trade balance is 0
- But nothing guarantees that $Y = Y_{TB}$

EQUILIBRIUM: $Y = Z$

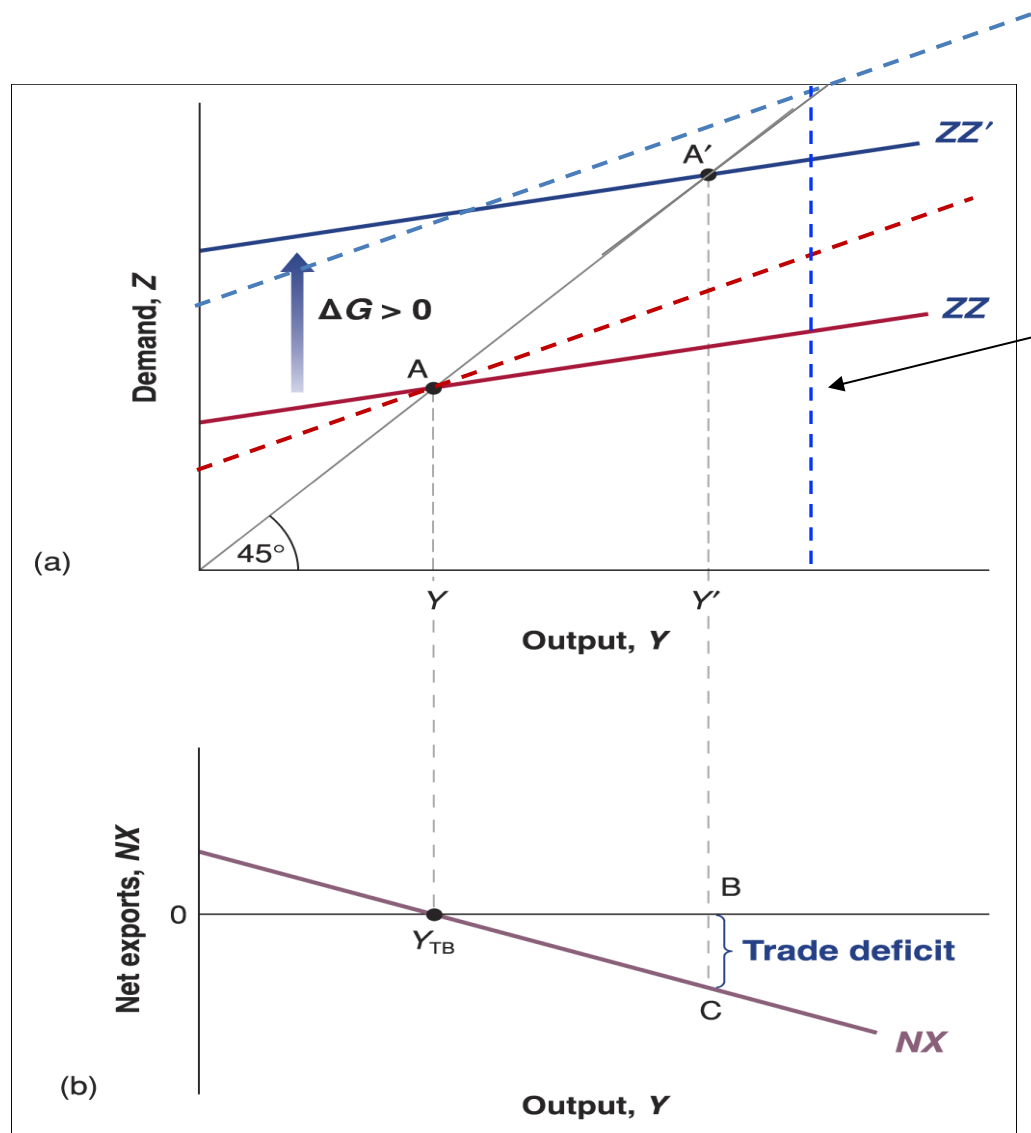
$$Y = C(Y - T) + I(Y, r) + G - IM(Y, \epsilon)/\epsilon + X(Y^*, \epsilon)$$



Increase in domestic demand in an open economy (for instance, increase in G)

- Assume economy in a recession, so government decides to increase G
- Assume that initially $Y = Y_{TB}$
- Start from the goods market equilibrium:
 - Increase in G moves ZZ up (by ΔG)
 - Y goes up, but less than in a closed economy, since part of higher domestic demand goes to imports!
 - **Multiplier is smaller in an open economy**
- What about TB ?
 - The increase in IM (for fixed X) leads to a **trade deficit**

Graphical effects of an increase in G

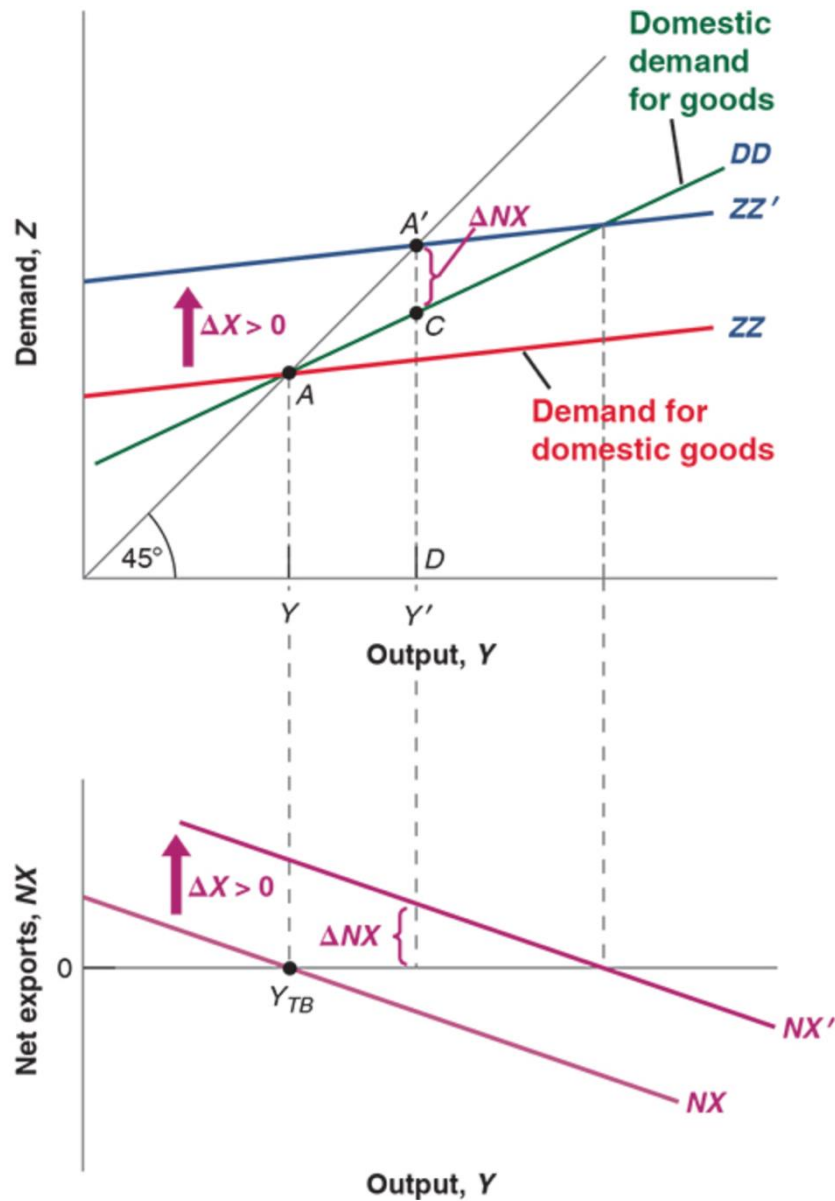


In a closed economy,
 Y would have
increased more!

Increase in foreign demand in an open economy (increase in Y^*)

- The increase in Y^* could for instance come from an increase in G^* of an important commercial partner
- Assume again that initially $Y = Y_{TB}$
- Start from the goods market equilibrium:
 - Increase in Y^* moves ZZ up (by ΔX)
 - Y goes up
- What about TB?
 - The increase in X shifts the NX curve up (by ΔX). Intuition: if X goes up, TB is obtained for a higher value of IM
 - What happens to the TB in the new equilibrium?
 - Maybe increase in IM (because Y is higher) is larger than increase in X , so that there is a trade deficit
 - We know this is not the case! There must be a trade surplus now. Why?
 - Because DD does not move!

Graphical effects of an increase in Y^*



Increases in Demand – Domestic or Foreign

So far, we saw that:

1. An increase in G leads to a **higher Y** , and a deterioration of the trade balance. Bad, as the country accumulates debts wrt the world. This may imply high interest rates....
2. An increase in G^* (and Y^*), leads to a **higher Y** and the trade balance improves.

Implications:

- Shocks to demand in one country affect all other countries (think about the 2008 recession).
- These economic interactions complicate the task of policy makers. **Policy coordination** is not so easy to achieve.
 - Suppose that two countries are in a recession.
 - Each country has an incentive to wait for the other country to increase G ! This improves output and generates a trade surplus!

The effects of a nominal depreciation

Since in the short-run prices are constant, a nominal depreciation leads to a real depreciation

A depreciation (decrease in ε) leads to:

- A higher X (domestic goods are cheaper), higher NX
- A lower IM (foreign goods are more expensive), higher NX
- An increase in $1/\varepsilon$ (imported goods cost more), lower NX

- The first two effects lead to a **higher** NX
- However, the third effect leads to a **lower** NX

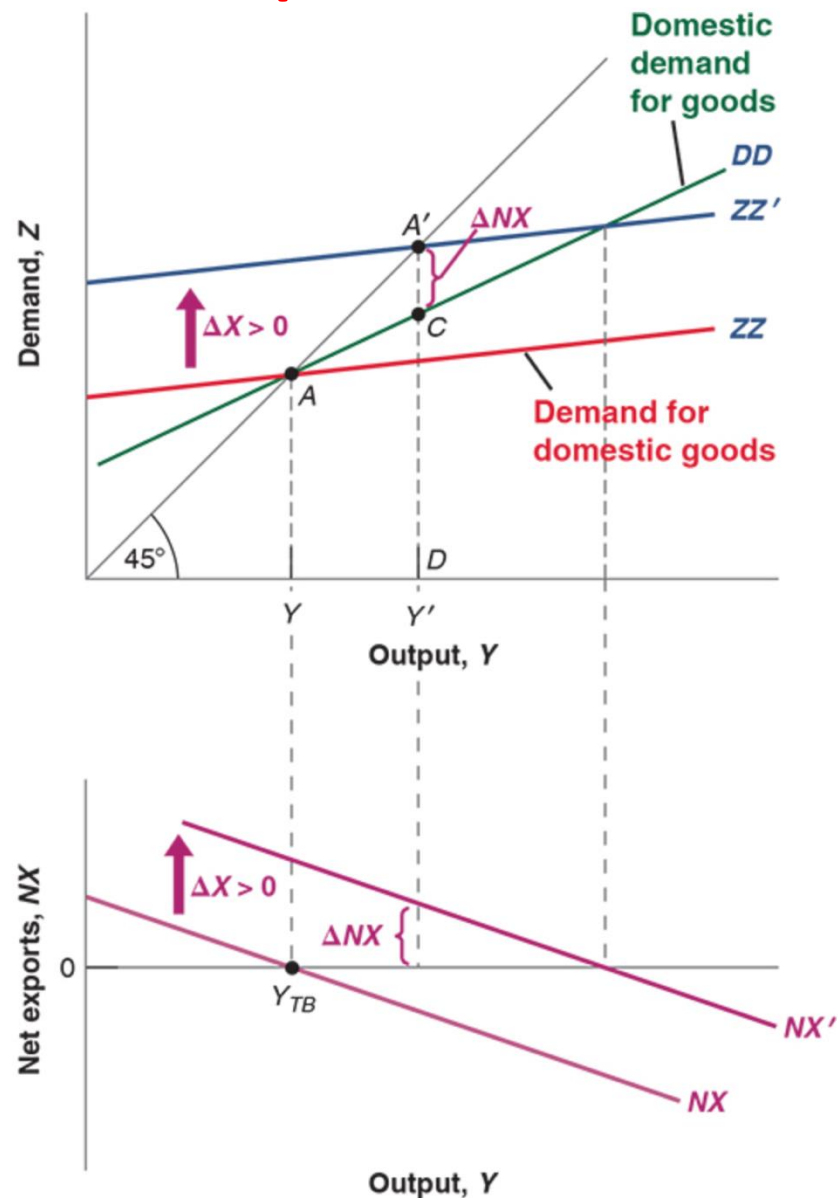
- Which one prevails? Data say that usually the first two effects are stronger, so:

A depreciation leads to an increase in NX (Marshall-Lerner condition)

Graphical effects of a nominal depreciation

A lower ε has the same effect as a higher Y^*
(so we use the figure we used before)

- Depreciation makes domestic goods cheaper for foreigners: increases X .
- Depreciation makes foreign goods more expensive for domestic consumers
- Given their incomes, domestic consumers are «poorer» given that wages take time to adjust (they pay more the same amount of foreign goods).
- This creates problems. Citizens are worse off.



Combining exchange rate and fiscal policy: policy mix

Consider the case of the following country:

- $Y = Y_n$ (natural level)
- but with a big trade deficit ($NX \ll 0$)

What can the government do to increase NX , without changing Y ?

- Decrease ε (depreciation)? NX improves, but Y increases
- Decrease G ? NX improves, but Y decreases

Combining exchange rate and fiscal policy: policy mix

The government can do both things at the same time:

1. Depreciation (shifts the curve NX)



Economy goes from A to A'

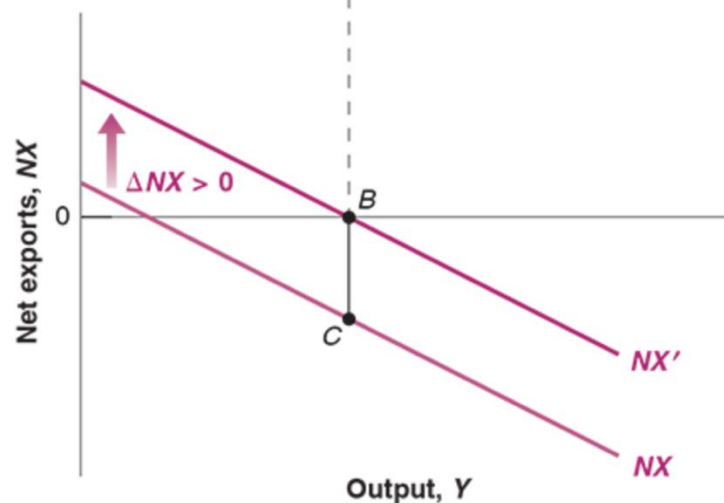
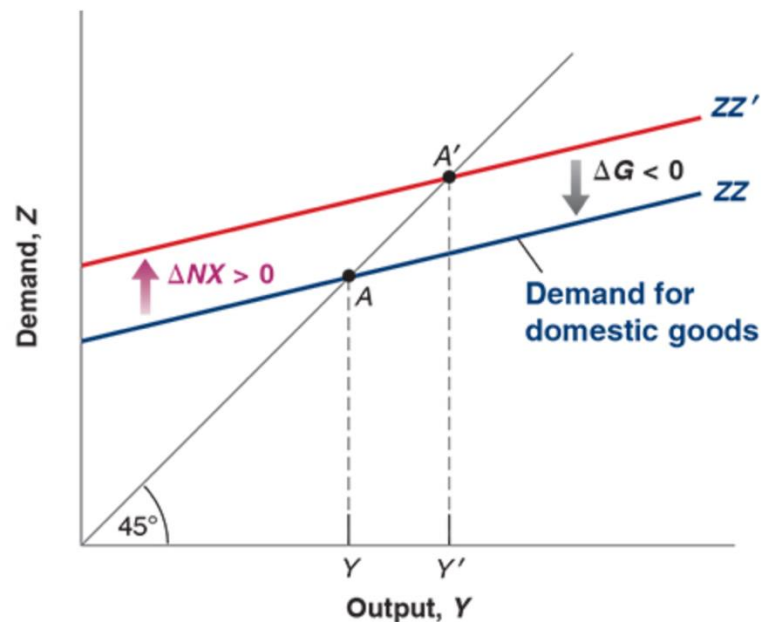
(higher NX, but $Y > Y_n$)

2. Lower G (does not change the curve NX)



Economy goes back to A

(NX=0 and $Y = Y_n$)



2 targets → 2 instruments

Exchange Rate and Fiscal Policy Combinations

Initial Conditions	Trade Surplus	Trade Deficit
Low output	$\epsilon? G\uparrow$	$\epsilon\downarrow G?$
High output	$\epsilon\uparrow G?$	$\epsilon? G\downarrow$

Focus on low output + trade surplus.

Two targets: we want to increase Y and decrease trade surplus .

$G \uparrow$ obtains both targets.

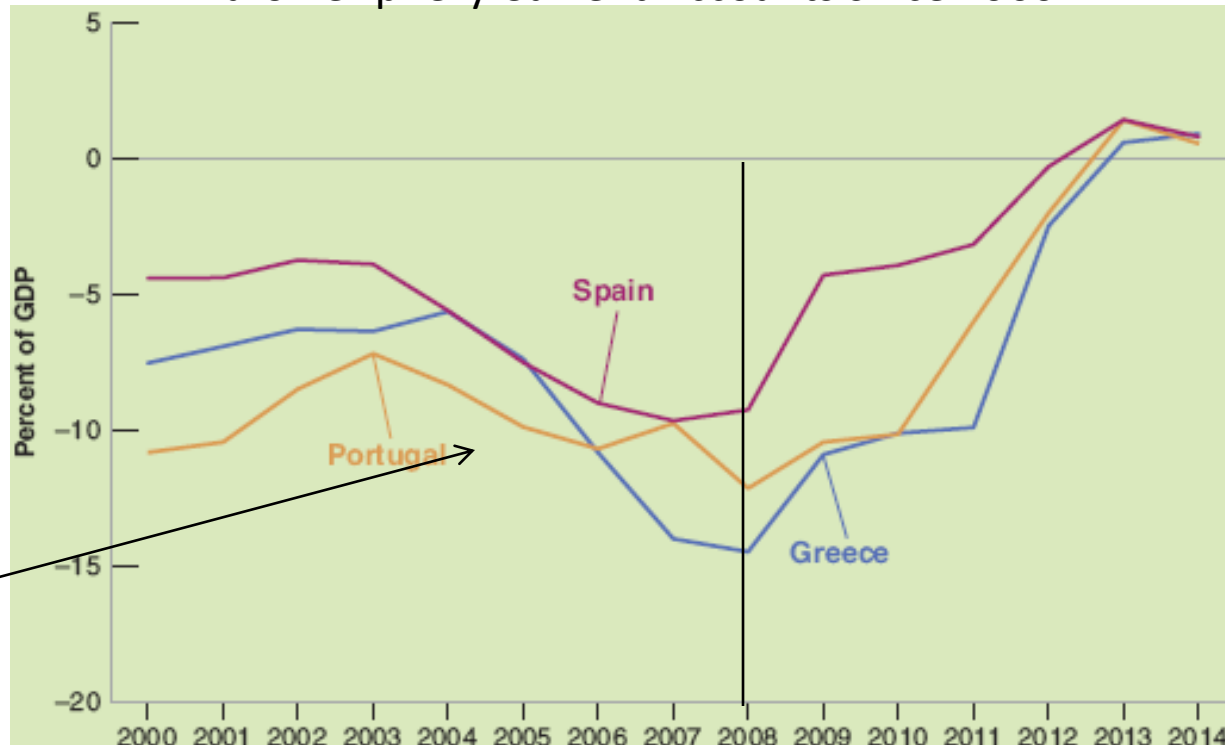
Variations in ϵ depend on the details of the economy: a depreciation or an appreciation (or no change) could be needed, since nothing ensures that $Y_n = Y_{TB}$

FOCUS: The Disappearance of Current Account Deficits in Greece: Good News or Bad News?

Reduction in current account deficits. Why?

1. country more competitive. ϵ decreases. Exports increase and imports decrease.
2. output decrease for a recession. Imports decrease. CA improves.

Euro Periphery Current Accounts since 2000

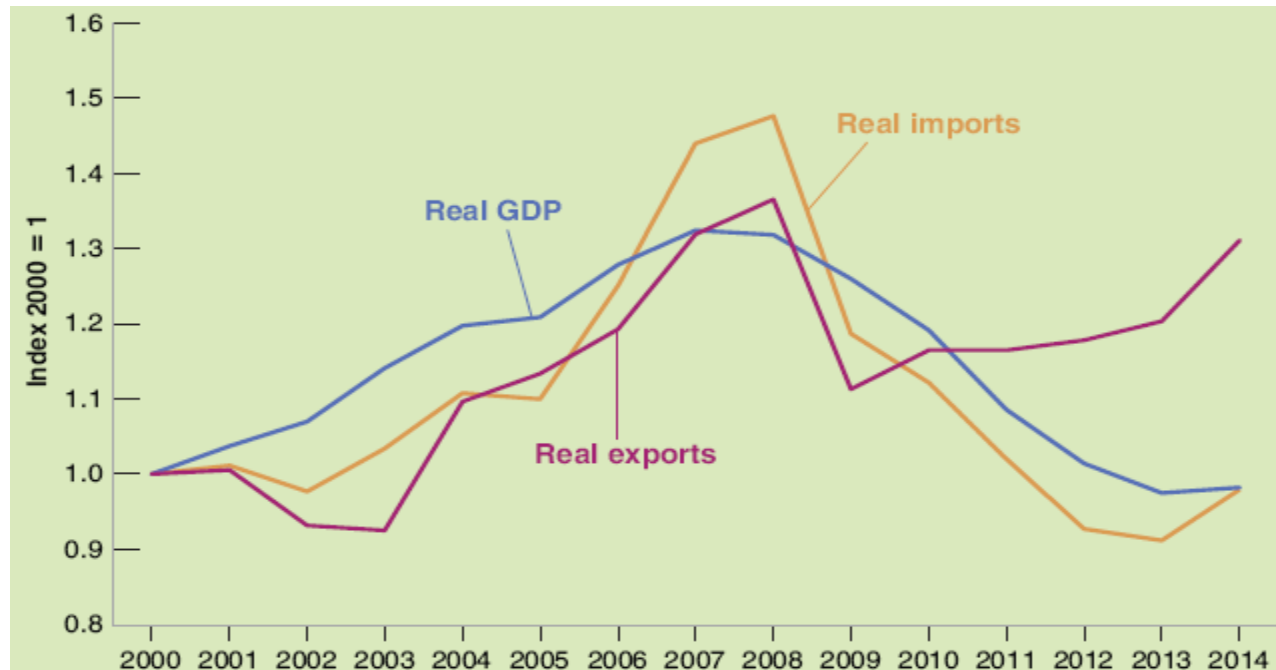


CA deficits!
KA surplus.

FOCUS: The Disappearance of Current Account Deficits in Greece: Good News or Bad News?

- You may want to adjust the nominal exchange rate to vary the real exchange rate. Euro. Impossible to change E . How to reduce ε ? By changing P : very hard... more on this later... this happened at a cost of reducing wages.
- Greece's real GDP and imports declined after 2008.
Import compression: A decrease in imports (improved current account balance) triggered by a decrease in output.

Imports, Exports and GDP in Greece since 2000



Saving/Investment and the trade balance in open economy

In a closed economy, we know that $S = I$

In an open economy, things change a bit:

$$Y = C + I + G + NX$$

⇓ Subtracting T from both sides

$$Y - T - C = I + G - T + NX$$

⇓ Adding net income (NI) to both sides

$$(Y - T + NI) - C = I + G - T + NX + NI$$

Disposable income

⇓

CA

$$CA = S + (T - G) - I$$

Saving/Investment and the trade balance in open economy

$$CA = S + (T - G) - I$$

- ❑ In open economy, it is no longer true that *total saving = investment*
- ❑ An excess of savings ($S + (T - G) > I$) corresponds to a **trade balance surplus: the country lends to the rest of the world** (our intuition from before!).
- ❑ An excess of investment ($S + (T - G) < I$) corresponds to a **trade balance deficit: the country borrows from the rest of the world** (our intuition from before!).
- ❑ If a country increases I , either it saves more or its CA deteriorates
- ❑ If government budget deteriorates, either S increases, or I decreases, or the CA deteriorates
- ❑ A country that saves a lot must either invest a lot or have a large CA surplus

Example: In China: $I < S$, and so $CA > 0 \rightarrow$ China «imports» assets ($KA < 0$).